

Course Handout

Institute/School	Chitkara University Institute of Engineering and Technology		
Department	Department of Computer Science & Engineering		
Programme	Bachelor of Engineering (B.E.), Computer Science & Engineering		
Course Title	Fundamentals of Computer Science	Session	2026-2027
Course Code	26CSE0107	Semester/Batch	1st/ 2026
L-T-P (Per Week)/No. of Hours (Per Week)	3-0-2/6	Course Credits	4
Pre-requisite	N/A	NHEQF Level	4.5
Course Coordinator	Dr. Mohammad Shuaib Khan	SDG Number	4,9

1. Scope and objectives of the Course

- To develop a fundamental understanding of computer systems, their organization, and core computing concepts.
- To build foundational knowledge of computer hardware, software, memory, and operating systems.
- To understand the role of programming languages, software development practices, and system software in modern computing.
- To understand the principles of operating systems, command-line environments, and essential system software used in computing.
- To understand the principles of software development, version control, and collaborative computing practices.
- To gain an overview of emerging computing paradigms, including cloud, distributed, and edge computing technologies.

2. Course Outcomes (COs)

CO1: Explain the fundamental concepts, characteristics, applications, and evolution of computers, including different generations of computers and their impact on modern society.

CO2: Analyze the functional components of a computer system, including CPU, memory units, input/output devices, buses, and the interaction among various hardware components.

CO3: Understand and classify different types of memory and storage technologies, including memory hierarchy, RAM, ROM, HDDs, SSDs, NVMe, and flash-based storage systems.

CO4: Identify various types of software, programming languages, language translators, and operating system functionalities, and demonstrate basic Linux command-line operations and cloud-based command execution

CO5: Apply fundamental concepts of software engineering practices, version control using GitHub, and modern computing environments such as cloud, distributed, and edge computing with virtualization and cloud service models (IaaS, PaaS, SaaS).

CO-PO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	No. of Lectures	NHEQF Descriptors
CO 1	3	2	–	–	1	–	–	–	–	2	–	2	1	10	Q1, Q4
CO 2	3	3	2	2	2	–	–	–	–	–	–	3	2	12	Q1, Q2, Q3
CO 3	3	2	1	1	2	–	–	–	–	–	–	3	2	10	Q1, Q2, Q3
CO 4	3	2	2	1	3	–	–	–	1	2	–	3	3	18	Q1, Q2, Q3, Q4
CO 5	2	2	3	2	3	1	–	–	3	2	2	3	3	25	Q2, Q3, Q4, Q6
Total Contact Hours 75															

H=High, M=Medium, L=Low

3. Recommended Books/E-Resources (Textbooks and Reference books):

- B01: Balagurusamy, E. Fundamentals of Computers (1st Edition). McGraw Hill Education, 2019.
 B02: Rajaraman, V. Fundamentals of Computers (6th Edition). PHI Learning, 2014.
 B03: Thareja, Reema. Fundamentals of Computers (2nd Edition). Oxford University Press, 2019.
 B04: Goel, Anita. Computer Fundamentals. Pearson Education, 2010.

B05: Mall, Rajib. Fundamentals of Software Engineering (5th Edition). PHI Learning, 2018

E-Library Resource

<https://library.chitkara.edu.in/subscribed-books.php>

4. Recommended Tools and Platforms

Ubuntu Linux (22.04 LTS or later) – Linux operating system for command-line and system programming.
GCC (GNU Compiler Collection) – C program compilation and execution.
GNU Make (Makefile) – Build automation and program execution.
Git & GitHub – Version control and collaborative software development.
Visual Studio Code (VS Code) – Source code editor with C/C++ and Git extensions.

5. Delivery/Instructional Plan and Resources

Theory Plan:

Lect. No.	Topics	CO	Book No, CH No, Page No	ILUM	Web References/Audio-Video
1-2	Introduction to Computers: Characteristics, Applications in daily life (Education, Healthcare, Banking, Engineering)	CO1	B01, Ch. 1, pp. 1-15	Lecture	https://nptel.ac.in/courses/106108101 https://www.youtube.com/watch?v=vBURt97EkA
3-4	Evolution and Generations of Computers (Generations I-V)	CO1	B01, CH1, Page No 1-17 B02, CH2, Page No 19-24 B03, CH 03, Page No 83-124	Discussion, Demonstration method, Questioning	https://nptel.ac.in/courses/10610521 https://nptel.ac.in/courses/10610521
5	Von Neumann architecture	CO1	B02, CH 6, Page No 117-149 B03, CH 04, Page No 137-179	Lecture	https://www.youtube.com/watch?v=TQWERtMoKbI
6-7	Block Diagram of a Computer System: Functional units—input, output, memory, CPU	CO1	B01, Ch. 3, pp. 35-52	Discussion, Demonstration method, Questioning	https://sites.ualberta.ca/~gingrich/courses/phys395/notes/node173.html
8-9	Description of data bus, address bus, control bus; component interaction.	CO2	B01, CH4, Page No 81-98 B02, CH9, Page No 237-280	Lecture	https://www.tutorialspoint.com/computer_organization/computer_organization_bus_structure.htm
10-12	Memory Hierarchy and Classification: Memory layers from registers to hard disk; volatile vs non-volatile memory;	CO2	B02, CH 10, Page No 283-306 B04, CH 08, Page No 238	Lecture	https://nptel.ac.in/courses/106108101 https://www.youtube.com/watch?v=LKYKp_ZzlvM https://www.youtube.com/watch?v=d9WyerblWQe&list=PLBlnK6fEYqRiVhbXDGLXdk_OQAeuVcp2O&index=74
CAT 1					
13-16	RAM types (SRAM, DRAM) and ROM types (PROM, EPROM, EEPROM).	CO3	B02, CH8, Page No 207-235	Lecture	https://www.geeksforgeeks.org/what-is-rom-read-only-memory/
17-19	Secondary storage Ecosystem, HDD Vs SSD, Flash media	CO3	B03, CH7, Page No 289-307	Lecture	https://www.ibm.com/think/topics/software
20-22	Types of Software and Programming languages, System software, application software, middleware	CO4, CO5	B03, CH 12, Page No 480-494	Discussion, Demonstration method, Questioning	https://www.geeksforgeeks.org/types-of-software/
23-25	Firmware; evolution from machine to high-level languages, Assembler, Compiler, Interpreter	CO5, CO4	B03, CH 09, Page No 400-426 B03, CH 10, Page No 436-447	Lecture	https://www.tutorialspoint.com/computer_programming/computer_programming_introduction.htm
CAT 2					



26-27	Introduction to Operating Systems: Different types of operating systems, evolution of operating system. Generations of operating system	CO2	B05, CH8, Page No 181-198 B04, CH9, Page No 237-280	Discussion	https://www.tutorialspoint.com/operating_system/os_overview.htm
28-29	Command Line Interfaces (CLI) and Basics terminal commands	CO4	B02, CH7, Page No 211-234	Discussion, Demonstration method, Questioning	https://aissmospoly.org.in/wp-content/uploads/2020/01/File-management-.pdf https://www.youtube.com/watch?v=DYGIPIez5GY&list=PLYSKRj7KSLYgS9jVyonZTFpiYkIzZXF2-
30-31	Introduction to terminal environments and essential navigation commands, File manipulation, Linux and Unix based permission settings	CO4	B02, CH3, Page no 61-67 B03, CH5, Page No 134-154	Discussion, Demonstration method, Questioning	https://codemaniacstudio.com/unit-1-introduction-to-linux-operating-system-2/ https://www.youtube.com/watch?v=h_xSovNstSE
32-33	Execution of Linux commands on Cloud	CO4	B02, Ch. 3, pp. 91-109	Discussion, Demonstration method, Questioning	https://training.linuxfoundation.org/resources/linux-basics/ https://docs.redhat.com/en/documentation/red_hat_enterprise_linux
CAT 3					
34-35	Software Engineering Lifecycle Foundations: Overview of the Software Development Life Cycle	CO5	B03, Ch. 5, pp. 234-244	Lecture	https://www.tutorialspoint.com/sdlc/sdlc_overview.htm
36-37	Software Development Life Cycle (SDLC) models	CO5	B01, Ch. 3, pp. 139-141	Lecture	https://www.javatpoint.com/software-engineering-sdlc-models
38	GitHub and version control system using github.	CO5	B01, Ch. 4, pp. 165-175	Discussion, Demonstration method, Questioning	https://docs.github.com/en/get-started/using-git/about-git
39-40	Overview of contemporary computing environment	CO4, CO5	B03, Ch. 3, pp. 139-141	Lecture	https://csrc.nist.gov/publications/detail/sp/800-145/final
41-42	Cloud, Distributed, and Edge computing networks	CO4, CO5	B02, Ch. 4, pp. 299-320	Discussion, Demonstration method, Questioning	https://azure.microsoft.com/en-us/resources/cloud-computing-dictionary/what-is-edge-computing/
43	Cloud Computing Frameworks & Decoupled Topologies	CO5	B02, Ch. 4, pp. 299-320	Lecture	https://www.redhat.com/en/topics/microservices
44	cloud deployment setups (IaaS, PaaS, SaaS)	CO5	B02, Ch. 4, pp. 299-320	Lecture	https://www.ibm.com/think/topics/virtualization
45	Virtualization basics	CO5	B02, Ch. 4, pp. 299-320	Lecture	https://www.ibm.com/think/topics/virtualization
CAT 4 & CAT 5					

Lab Plan:

Lab No.	Experiment	CO	TLM	Web References/Audio-Video
1-2	a) Installation: Configuration & Customisations of Linux b) Introduction to GCC compiler: Basics of GCC, Compilation of program, Execution of program c) Time stamping, Automating the execution using makefile.	CO4	Discussion, Demonstration method, Questioning	https://ubuntu.com/tutorials/install-ubuntu-server#1-overview https://www.youtube.com/watch?v=oLjN6jAg-sY
3-4	Basic Commands of Linux OS: Navigation (ls, cd), File manipulation (cp, mv, rm), and Viewing (cat, less), where is, what is and their options	CO4	Discussion, Demonstration method, Questioning	http://tldp.org/LDP/abs/html/basic.html https://www.youtube.com/watch?v=YHFzr-akOas
5-6	Execute Directory-oriented commands: cd, pwd, mkdir, rmdir and their options	CO4	Discussion, Demonstration method, Questioning	http://litux.nl/Reference/Books/7213/du0082.html https://www.youtube.com/watch?v=YHFzr-akOas
CAT 1				
7-8	Execute the basic and user status commands like: su, sudo, man, help, history, who, whoami, id, uname, uptime, free, tty, cal, date,		Discussion, Demonstration	https://www.geeksforgeeks.org/linux-unix/basic-linux-commands



	hostname, reboot, clear and their options	CO4	method, Questioning	
9-10	Write programs to create and manage processes using system calls such as fork(), exec(), and wait(). Study parent-child relationships and process lifecycle behaviour in Linux.	CO4	Discussion, Demonstration method, Questioning	https://iies.in/blog/understanding-fork-exec-and-process-creation-in-linux/
11-13	Implement Linux memory management commands: Overview, free, vmstat, Memory Allocation System Calls (malloc, sbrk). Analyse how memory is allocated and utilized by running processes.	CO2, CO4	Discussion, Demonstration method, Questioning	https://linuxhint.com/fork-system-call-linux/
CAT 2				
14-16	To understand the basic concepts of version control and perform fundamental Git operations such as repository creation, staging, committing, and viewing commit history.	CO5	Discussion, Demonstration method, Questioning	About-Version-Control">https://git-scm.com/book/en/v2/Getting-Started>About-Version-Control
17-19	To learn and execute basic Linux command-line operations for file management, directory navigation, file viewing, and obtaining command help through the terminal.	CO4	Discussion, Demonstration method, Questioning	https://minnie.tuhs.org/CompArch/Labs/week9.html https://www.cse.iitb.ac.in/~mythili/os/iitb_slides/memory-syscall.pdf
20-21	To create and manage a GitHub repository and synchronize local and remote repositories using basic Git commands.	CO5	Discussion, Demonstration method, Questioning	https://docs.github.com/en/repositories/creating-and-managing-repositories/creating-a-new-repository
CAT 3				
22-23	To understand the basic concepts of cloud computing, including cloud service models (IaaS, PaaS, SaaS) and deployment models through hands-on activities.	CO5	Discussion, Demonstration method, Questioning	https://aws.amazon.com/types-of-cloud-computing/
24-25	To create and configure a cloud-based virtual machine and perform basic Linux operations on the virtual machine.	CO4, CO5	Discussion, Demonstration method, Questioning	https://cloud.google.com/compute/docs/instances/create-start-instance
26-28	To understand the concept of virtualization by creating and using a virtual machine in a virtualization environment.	CO5	Discussion, Demonstration method, Questioning	https://www.vmware.com/topics/glossary/content/virtualization.html
29-30	To analyze computer system performance by monitoring CPU utilization, memory usage, disk activity, and running processes using Linux system monitoring command such As top, htop, vmstat, iostat and free	CO2, CO4	Discussion, Demonstration method, Questioning	https://linuxcommandlibrary.com/man/top https://man7.org/linux/man-pages/man1/vmsta
CAT 4 & CAT 5				

6. Course ERISE Mapping:

Skill Enhancement
Employability

7. Active Learning Methodology Used

These are learner-centered teaching methodologies that promote active participation, critical thinking, and meaningful knowledge acquisition. Group discussions encourage collaborative learning through the exchange of ideas, perspectives, and problem-solving among students. Student-created presentations and visual learning aids, such as PPTs, charts, matrices, flowcharts, and models, foster creativity, research skills, conceptual understanding, and effective communication by enabling learners to organize and present information systematically. Quiz and test questions serve as formative and summative assessment tools that evaluate learning outcomes, reinforce knowledge retention, provide timely feedback, identify learning gaps, and support continuous improvement in students' academic performance and overall competency development.

- Group Discussion
- Student-Created PPT, Charts, Matrices, Flowcharts, Models
- Quiz/Test Questions

8. Action plan for different types of learners

Fast Learners	Average Learners	Learners with constant monitoring
- Group discussion - Quiz/competition - Seminars	- Workshops - Formative Exercises used to highlight concepts and notions - E-notes and E-exercises to read ahead of the pedagogic material.	- Remedial Classes on Saturdays - Encouragement for improvement using Peer Tutoring - Use of Audio and Visual Materials - Use of Real-Life Examples

9. Additional Self-directed Learning

Assignments to promote self-learning, survey of contents from multiple sources.

S.No	Advanced Topics, Additional Reading, Research papers and any	CO	ALM	References/MOOCs
1.	Mobile Operating Systems Features of Android and iOS	CO2	Think – Pair- Share	https://www.geeksforgeeks.org/operating-systems/difference-between-ios-and-android/
2.	Cloud Computing Cloud OS Internals: Resource Virtualization in AWS, Azure	CO5	Shadowing	https://www.youtube.com/watch?v=cmdTnA6hN_A

10. Delivery Details of Content Beyond Syllabus

Content beyond syllabus covered (if any) should be delivered to all students that would be planned, and schedule notified accordingly. You can include industry intervention session under it.

- Binary data processing in digital devices
- From idea to software product: SDLC in industry

11. Evaluation Scheme & Components:

Assessment Type	Evaluation Component	No. of Assessments	Type of Component	Component Description	% Weightage of Component	Max. Marks	Mode of Assessment	CO
Formative	Component1	01	Continuous Assessment(CE)	CAT1	25%	25	Computer Based Test	CO1, CO2
Formative	Component2	01	Continuous Assessment(CE)	CAT2	25%	25	Computer Based Test	CO2, CO3
Formative	Component3	01	Continuous Assessment(CE)	CAT3	25%	25	Computer Based Test	CO3, CO4, CO5
Formative	Component4	01	Continuous Assessment(CE)	CAT4	25%	25	Computer Based Test	CO1, CO2, CO3, CO4-CO5
Formative	Component5	01	Continuous Assessment(CE)	CAT5	25%	25	Computer Based Test	CO1, CO2, CO3, CO4-CO5
Total				100%				

Note:

- The academic performance for each subject is evaluated across five Continuous Assessments (CA1 through CA5).
- Best-of-Four Policy: The final score will be calculated based on the best four scores out of the five assessments.
- Mandatory Requirement: Students must appear in at least one of CA4 or CA5. Failure to appear in both CA4 and CA5 will result in an Incomplete (I) grade being awarded for the respective subject.
- Passing Threshold: Students must secure a minimum aggregate of 50% marks to pass a subject. Scoring below 50% will result in a Fail grade ('F').

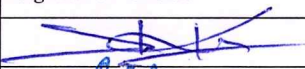
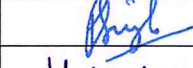
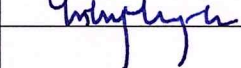

12. Sessional Test-Wise Syllabus of the Course:

Evaluation Component	Description	Syllabus Covered	Weightage %
Component1	CAT1	Lecture (1-12)	25%
Component2	CAT2	Lecture (13-25)	25%
Component3	CAT3	Lecture (26-33)	25%
Component4	CAT4	Lecture (34-45)	25%
Component5	CAT5	Lecture (34-45)	25%

13. Academic Integrity Policy:

Education at Chitkara University builds on the principle that excellence requires freedom where Honesty and integrity are its prerequisites. Academic honesty in the advancement of knowledge requires that all students and Faculty respect the integrity of one another's work and recognize the importance of acknowledging and safeguarding intellectual property. Any breach of the same will be tantamount to severe academic penalties.

This Document is approved by:

Designation	Name	Signature with date
Course Coordinator	Dr. Mohammad Shuaib Khan	
Program in Charge	Dr. Preetinder Singh Brar	
Pro. Vice Chancellor	Dr. Jaiteg Singh	
Date(DD/MM/YYYY)	19/08/2026	